NoisePrivacyMap<L2Distance<RBig>, MultiDP> for ZExpFamily<2>

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of NoisePrivacyMap for ZExpFamily<2> with output measure MultiDP.

1 Hoare Triple

Precondition

Compiler-verified

NoisePrivacyMap is parameterized as follows:

- MI is L2Distance<RBig>.
- MO is MultiDP.

Caller-verified

None.

Pseudocode

```
class ZExpFamily2:
    def noise_privacy_map(
        self, _input_metric: L2Distance[RBig], _output_measure: MultiDP
    ) -> PrivacyMap[L2Distance[RBig], MultiDP]:
        zcdp_map = self.noise_privacy_map(  #
            L2Distance.default(), zCDP
        )

        def privacy_map(d_in: RBig):
            rho = zcdp_map(d_in)
            return PrivacyGuarantee().with_zCDP(rho, 0.0)  #

        return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, the implementation returns an error if `self` is invalid. Otherwise it returns a privacy map for the mechanism `function(x) = x + Z`, where the coordinates of Z are independent discrete Gaussian samples represented by `self`, under `MultiDP`.

Proof. Line 5 delegates to the exact-zCDP privacy map proved in `NoisePrivacyMap_for_ZExpFamily2.tex`. That map validates the scale and sensitivity and returns the conservative value of ρ established by the discrete-Gaussian zCDP theorem.

Line 11 wraps this exact zCDP distance in a `PrivacyGuarantee` with source delta zero. Consequently the advertised `MultiDP` capability contains exactly the zCDP representation already proved for this mechanism. No Gaussian-DP or RDP representation is inferred from the distribution name; those representations, if useful, can be derived later by consumers using their respective theorems. The privacy guarantee is therefore valid under `MultiDP`, and increasing the distance to any `d_out` accepted by the measure ordering preserves the guarantee. □

References